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Impact of Changes in Marriage Law

Implications for Fertility and School Enrollment

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ABSTRACT

Does the postponement of marriage affect fertility and investment in human capital? I study this question in the context of a 1957 amendment to the Mississippi marriage law that was aimed at delaying the age of marriage; changes included raising the minimum age for men and women, parental consent requirements, compulsory blood tests, and proof of age. Using a difference-in-differences design at the county level, I find that, overall, marriages per 1,000 in the population in Mississippi and its neighboring counties decreased by nearly 75 percent; the crude birth rate decreased between 2 and 6 percent; and school enrollment increased by 3 percent after the law was enacted (by 1960). An unintended consequence of the law change was that illegitimate births among young black mothers increased by 7 percent. I show that changes in labor market conditions during this period cannot explain the changes in marriages, births, and enrollment. I conclude that stricter marriage-related regulations that lead to a delay in marriage can postpone fertility and increase school enrollment.

I. Introduction

The decision of when to marry has important consequences for men and women. Particularly for women, early marriage is often associated with lower socioeconomic status and less schooling (Dahl 2009, Field and Ambrus 2006). Marital status is also known to be an important determinant of female labor force participation (Angrist and Evans 1996, Heckman and McCurdy 1980, Stevenson 2008); moreover, women seem to invest more in their careers if they delay fertility and marriage (Goldin and Katz 2002). It is clear that women's (and to some extent men's) marital decisions are intricately tied to their economic outcomes. While marriage is considered a choice, laws regard-

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ing marriage often control various aspects of who marries, when people marry, partner choice, and number of partners. Moreover, societal norms place importance on the act of marriage to legitimize cohabitation and childbearing. In the United States, for example, married couples form 90 percent of all heterosexual couples (U.S. Census Report 2007),¹ and the majority of children are born to married couples (Hamilton et al. 2005). Given the central role of marriage (78 percent of all women older than 18 marry), marriage laws can have direct implications for the economic outcomes of men and women.

Marriage laws can be used as a policy tool as well—in 1980, China raised the age of marriage for women in a bid to control fertility. On the other hand, if legal marriage is just a formality or if marriage laws are routinely ignored,² then it is likely that changes in marriage law will not have much of an impact. Given the intended policy goals of marriage laws as well as rising cohabitation rates, an important empirical question surrounding marriage laws is whether changes in marriage laws, particularly changes that raise the cost of marriage, have an impact on marriage rates, fertility, and schooling. Fertility and schooling reflect key investments that men and women make early in their adult lives that have long-term consequences for welfare and labor market outcomes; hence, from a policy perspective it is relevant to know whether and how marriage laws affect these investments. Although many recent papers have examined the consequences of divorce laws,³ few have studied the impact of marriage law changes on outcomes such as fertility and schooling. Dahl (2009) uses marriage law changes as an instrument for delayed marriage to study the relationship between early teen marriage and poverty. However, Dahl uses marriage law changes along with schooling changes to analyze the impact of both laws simultaneously as changes in compulsory schooling laws often coincide with changes in marriage laws. Buckles, Guldi, and Price (2009) examine the role played by blood test requirements for obtaining marriage licenses, and they find that even small increases to the cost of marriage can decrease marriage rates and increase the incidence of illegitimate births. Finally, Blank, Charles, and Sallee (2009) find evidence of misreporting of age in marriage licenses, and they also find that young men and women tend to avoid restrictive marriage laws by getting married in a different state. This paper adds to this body of work by examining not only the effect of changes in marriage laws on marriage rates but also on fertility and educational investments. While this paper examines the impact of a wide range of marriage law changes, the results make it clear that the age restrictions perhaps have the largest impact. Hence, this paper complements Buckles, Guldi, and Price (2009) in showcasing another aspect of marriage law changes and its impact on a wide range of policy relevant outcomes.

A priori, it is not clear what effect increasing barriers to marriage will have on fertility and schooling. If entering into marriage becomes harder, individuals simply may have children out of wedlock, exacerbating the problem since then the father is less

1. While rates of cohabitation have been on the rise in the United States, demographic evidence suggests that it is not becoming a substitute to marriage (Raley 2001).

2. A case in point is the change in marriage law in India in 1978 that raised the minimum age of marriage. The Child Marriage Restraint Act of 1978 increased the age of marriage for women from 15 to 18. However, the data shows no sharp breaks around this time period in the age at which women got married. Moreover, survey data evidence shows that awareness of these marriage laws is also weak in the Indian context.

3. Stevenson (2007, 2008), Stevenson and Wolfers (2006), and Rasul (2006) are some of the many papers that consider the impact of divorce laws on various outcomes like fertility, education, marriage quality, and incidences of domestic violence.

likely to help raise the child. After marriage, the sharing of resources becomes easier; hence, spouses may also be more likely to have a chance to get further education. So, the postponement of marriage could reduce education (Stevenson 2007 shows that divorce laws negatively impact spousal support for education-related investments). Alternatively, if people are reluctant to have children out of wedlock, postponement of marriage could lead to a drop in the birth rate. Moreover, unmarried women lacking spousal support might have more incentives to invest in their own education. Therefore, marriage laws could increase school enrollment and educational attainment; the impact of changes in marriage law is essentially an empirical question.

In 1957, the state of Mississippi amended its marriage law. Although the marriage law amendment was passed in 1957, it went into effect in 1958 and immediately resulted in a substantial decline in marriages performed in Mississippi (see Figure 1). The changes included an increase in the minimum marriage age for women from 12 to 15 years old and for men from 14 to 17 years old. The law also introduced a parental consent requirement if either party was younger than 18, a compulsory three-day waiting period, serological blood tests, and proof of age. In addition, brides under the age of 21 were required to marry in their county of residence. Hence, the barriers to marriage were raised not just via increasing minimum age laws but also by introducing blood tests and other requirements, which affected everyone. Using a difference-in-differences strategy that considers Mississippi and its surrounding counties as the treatment group and remaining counties in the states neighboring Mississippi (Alabama, Louisiana, Arkansas, and Tennessee) as the control group, I find that by 1960 marriages per 1,000 decreased by 75 percent, the crude birth rate (births per 1,000 in the population) decreased between 2 and 6 percent, and enrollment among 14–17-year-olds increased by 3 percent more than the corresponding change in the control group. Using yearly state-level data from 1952 to 1960, I find that the percentage of total births to young black women decreased by 8 percent; for white women, the decline was 18 percent more than the change in the control group. However, this decrease in births is mitigated by an increase in illegitimate births, primarily among black women.

I focus on short-run effects of the law change as there were tremendous social and economic changes in the 1960s—particularly due to the Civil Rights Act of 1964—that could confound an analysis of long-run outcomes. Moreover, the 1960s and 1970s saw the introduction of various contraceptives (the birth control pill in particular), the legalization of abortion, and changes in divorce laws, which could also affect marriage and fertility (Goldin and Katz 2002, Donohue and Levitt 2001, Ananat and Hungerman 2007, Bailey 2006). With this caveat in place, using the 1990 census, I do not find long-run effects on fertility, suggesting that the drop in birth rates is simply a delay in fertility.⁴ I do find that women affected by the law were more likely to complete high school in the long run.

This paper's relevance extends to the context of developing countries, where age at first marriage and educational attainment of women tend to be quite low. Therefore, for women in developing countries who have high levels of fertility and low levels of educational attainment, laws that delay marriage might be welfare improving. Field and Ambrus (2008) show that women who marry later attain more years of education in Bangladesh. Given this finding, they hypothesize that imposing universal age of consent laws can raise educational levels of women. The findings of this paper provide direct corroborating evi-

4. This is in contrast to an earlier version of the Field and Ambrus (2006) paper, Field (2004), where a delay in marriage is found to have a sizable but barely significant impact on completed fertility.

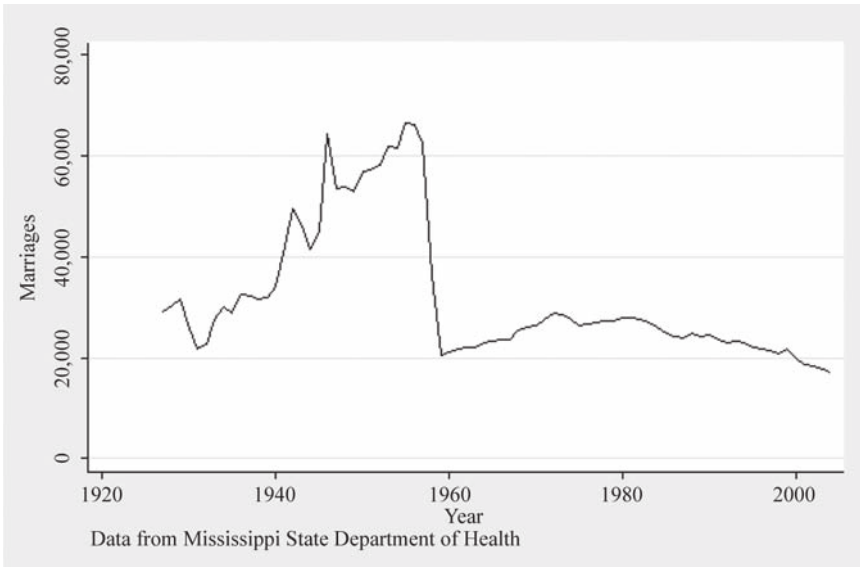


Figure 1
Total Number of Marriages by Year in Mississippi

Notes: Data from Vital Statistics of Mississippi. Figure shows number of marriages (regardless of residence of bride and groom) that occurred in Mississippi.

dence towards this idea, albeit in the setting of the American South. Marriage laws are also thought of as a tool for reducing fertility. The results of this paper suggest that raising the minimum age of marriage delays fertility but perhaps has little effect on completed fertility.

II. Background and Preliminary Evidence of Marriage Decline

In 1957, the state of Mississippi amended its existing marriage laws. A *Time Magazine* article from December 1957 carried a prediction that the proposed changes in marriage law would “shoot out loveland’s neon lights and keep rash child brides at home.” In particular, the article mentions that, due to lenient laws in Mississippi compared to its neighboring states, the state’s border counties were a haven for early marriages. Changes in Mississippi’s marriage laws were driven by “increased pressure from physicians, ministers and clubwomen.” The changes included raising the minimum age of marriage for women from 12 to 15 and for men from 14 to 17 as well as introducing parental consent laws. Circuit clerks were required to notify the parents of minors via registered mail during the mandatory three-day waiting period (which was also introduced at the same time). Age verification and blood tests were additional requirements that were added to the existing laws.

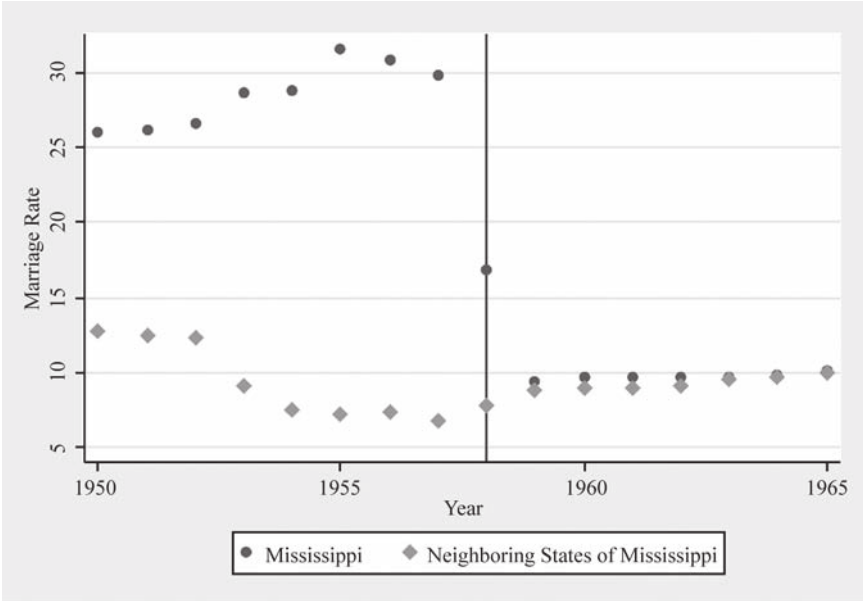


Figure 2
Marriage Rate in Mississippi and Surrounding States

Notes: Data from Vital Statistics. Neighboring states of MS are Arkansas, Tennessee, Louisiana, and Alabama. Marriage rates are marriages per 1,000 in the population.

Did these changes lead to a decline in marriages? To answer this question, I begin by summarizing some of the findings of Plateris (1966), who first examined the impact of this law change on marriage rates. I also present additional evidence of the marriage decline using data from the Vital Statistics and the City and County Handbooks (Section III describes the data used in this paper in detail). The evidence I present here is largely graphical and the econometric evidence is presented later in the paper.

Although the marriage law amendment was passed in 1957, it went into effect in 1958 and immediately resulted in a substantial decline in marriages performed in Mississippi. The resulting decline in Mississippi was a combination of a decline from out-of-state couples and in-state couples who did not meet the requirements. Indeed, Mississippi's laws prior to 1958 were lenient in comparison to its surrounding states of Alabama, Arkansas, Louisiana, and Tennessee. Figure 2 uses yearly marriage rates from the Vital Statistics to show the decline in Mississippi relative to its surrounding states. Importantly, it shows that, while marriages in Mississippi declined and marriages in surrounding states increased, in no way did the increase compensate for the massive decline. Plateris (1966) uses state of residence information from the Mississippi State Board of Health to confirm this (see Appendix Table 1).⁵ From 1957 to 1959, Plateris (1966) notes, marriages in Mississippi where both spouses resided in the

5. Appendix tables are available at <http://prbharadwaj.wordpress.com/papers/>.

state declined by 20 percent while marriages in Mississippi where both spouses were nonresidents of Mississippi declined by 90 percent. As a result, most of the decline in Mississippi itself was driven by the drop in out-of-state marriages. Out-of-state couples who did not get married in Mississippi could still get married in their home state and many did so. However, even taking the increase in marriages in neighboring states into account, the area in general saw a decline in the number of marriages by 13 percent. I confirm these findings using county-level marriage data and plotting changes in marriage rates between 1950–54 and 1954–60. Figures 3a and 3b clearly show a large decline after the law change in counties that were closer to the Mississippi border (in particular along the Mississippi-Alabama border).

As further evidence, Figure 4a shows the drop in marriages at the state level using census data from 1930 through 1970. This figure is also useful in seeing that before 1950 we do not see any Mississippi-specific trends in marriages. In fact, between 1930 and 1950, trends in marriage in Mississippi looked very similar to trends in its neighboring states as well as other Southern states.

After 1950, however, we see that the proportion of married 19-year-olds (they were 16–17 when the law was passed) in Mississippi drops sharply below that of all other states. Figure 4b uses age of marriage data from the 1960 and 1970 census to compute the fraction of married teenagers. While this figure is predictably noisier, the decline in marriage in Mississippi is apparent. Importantly, these graphs show why a difference-in-differences approach is needed. The proportion of married 19-year-olds, or the fraction of married teenagers in other states, also seems to have declined in the late 1950s; hence, it will be crucial to separate the decline due to the secular trend from the decline due to the marriage law.

III. Data and Empirical Strategy

Isolating the impact of the change in marriage law is a critical challenge in this case. It is likely that changes in schooling and fertility of women is caused by local and/or macroconditions unrelated to the change in marriage law. The main strategy used to differentiate the effect of the law from other causes in this paper is a difference-in-differences strategy; however, data limitations for certain outcomes and years determine the type of difference-in-differences strategy used (see Table 1). Critical for my strategy, neighboring states of Mississippi did not experience a change in marriage law during this period (1950–60), and no state considered for the analysis experienced a change in compulsory schooling laws. Moreover, Arkansas, Tennessee, and Louisiana's minimum marriage age for women was 16 while Alabama's was 14 during 1950–60. In fact, Mississippi was the only state in the country during this decade to have the minimum age at 12 for women — all states had a higher minimum age by 1950.⁶

One of the main challenges while using a difference-in-differences strategy is to appropriately define treatment and control groups. Section II suggests a few different

6. One might worry about migration out of Mississippi to get married after the change in law, especially to Alabama since the minimum age there was less than Mississippi after 1957. However, we know from Plateris (1966) that this was not very prevalent; he reports a net decline of 20 percent for Mississippi resident marriages.

Table 1
Baseline Comparisons (Means Computed from 1950 and 1954)

Variable	Treatment		Control		Difference (Treatment-Control)	Standard Error
	Mean	Standard Deviation	Mean	Standard Deviation		
Panel A - Treatment is Mississippi and neighboring counties, control is remaining counties in neighboring states						
Marriages per 1,000 in population	23.09	45.66	10.53	17.96	12.56***	2.29
Log crude birth rate	3.16	0.22	3.06	0.303	0.09***	0.022
Percent 14-17-year-olds enrolled	77.64	5.12	76.89	6.14	0.75	0.67
Manufacturing wage	2.78	9.33	4.11	12.77	-1.32	0.95
Percent employed in manufacturing	35.25	27.83	44.33	39.35	-9.08***	2.909
Farms per 100 in population	0.12	0.04	0.097	0.04	0.02***	0.002
Percent farms with tractors	27.22	8.56	33.12	13.38	-5.89***	1.37
Agricultural employment per 1,000 in population	150.93	58.9	115.11	53.92	35.82***	6.25
Panel B - Excluding Mississippi counties						
Marriages per 1,000 in population	19.97	44.7	10.53	17.96	9.44***	2.44
Log crude birth rate	3.15	0.213	3.06	0.3033	0.088***	0.027
Percent 14-17-year-olds enrolled	76.02	4.31	76.89	6.14	-0.86	1.23
Manufacturing wage	3.73	11.6	4.11	12.77	-0.376	1.19

Percent employed in manufacturing	36.96	29.174	44.33	39.35	-7.37**	3.57
Farms per 100 in population	0.11	0.04	0.1	0.04	0.016***	0.004
Percent farms with tractors	27.22	8.56	33.12	13.38	-5.89***	1.37
Agricultural employment per 1,000 in population	137.85	53.95	115.11	53.92	22.73**	10.7
Panel C - Excluding Mississippi counties and within 200 miles of Mississippi border						
Marriages per 1,000 in population	19.97	44.7	9.56	19.93	10.41***	3.13
Crude birth rate	3.154	0.213	3.13	0.321	0.022	0.03
Percent 14-17-year-olds enrolled	76.02	4.31	77.53	6	-1.51	1.23
Manufacturing wage	3.73	11.6	4	14.15	-0.269	1.38
Percent employed in manufacturing	36.96	29.17	41.16	33.97	-4.19	3.36
Farms per 100 in population	0.11	0.04	0.09	0.043	0.019***	0.004
Percent farms with tractors	27.22	8.56	36.44	13.38	-9.216***	1.59
Agricultural employment per 1,000 in population	137.85	53.95	117.63	53.29	20.21*	11.02



Figure 3a
Change in Marriages 1950–54



Figure 3b
Change in Marriages 1954–60

Notes: Change in marriages is the percent change from the previous year of available data. Absolute value of the percent change is shown on the maps. County-level data are from the County and City Datebooks 1950, 1954, and 1960.

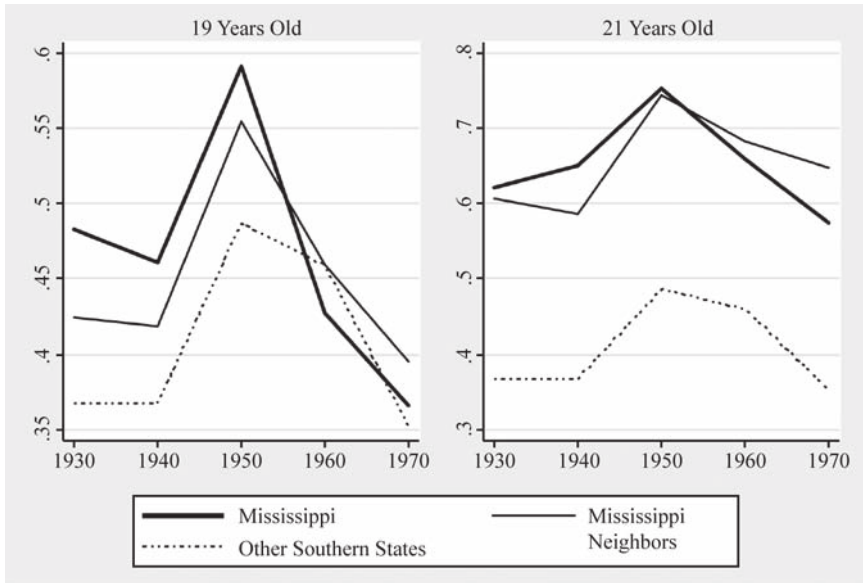


Figure 4a
Proportion of Married Women by Census Year

Notes: Data for Figure 4a uses the Census of 1930–70.

ways to define treatment and control groups. The definition of treatment is based largely on the degree of data disaggregation available. Using county-level data, we can use distance from the Mississippi border as a continuous measure of treatment or Mississippi and its surrounding counties as treatment and remaining counties in the neighboring states as control as a dichotomous measure of the areas that were affected by the law change (as suggested by Figure 3). Using state-level data, we can use Mississippi and neighboring states as treatment and remaining Southern states (where Southern is defined as a census region) as the control, for example. In addition, the nature of the law change implies that younger age groups should be more affected than older age groups. The age dimension of the law changes allows for a triple difference strategy (that is, comparing younger and older age groups across treatment and control states before and after the law change).

However, as I explain below, some of these strategies cannot be used in conjunction due to data limitations. Importantly, some of the data are available for more years, which allows for a more accurate control for trends prior to the law change, another aspect of a difference-in-differences analysis that is quite critical. The subsections below explain in detail the different strategies used.

A. County-Level Analysis

County-level analysis has the distinct advantage that I can examine changes in border counties relative to interior counties of neighboring states. Specifically, it allows me

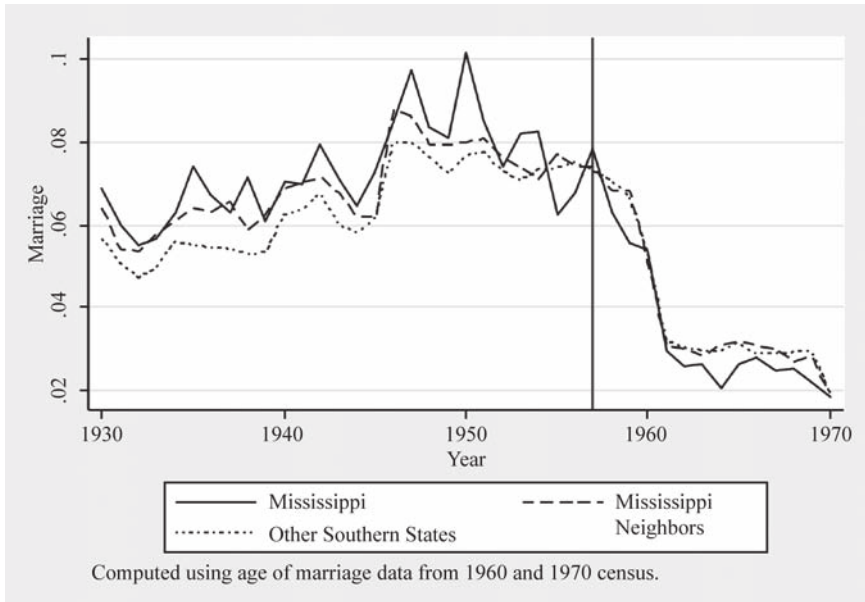


Figure 4b
Fraction of Married Teenagers by Year

Notes: Data for Figure 4b use the 1960 and 1970 Census and compute fraction married using data on age of marriage reported.

to construct treatment groups such that the intensity of “treatment” decreases with distance away from the Mississippi border. Comparing counties within the same state that only differ by distance to the Mississippi border reduces the possibility that factors other than the law change in Mississippi are driving the results. However, this strategy comes at the cost of not being able to include Mississippi as part of the results. To include Mississippi as part of the treatment group, I define a more restrictive treatment group comprising counties in Mississippi and counties in neighboring states that share a border with Mississippi. The remaining counties in neighboring states comprise the control group under this specification. Evidence from Section II suggests that this is a reasonable way to assign treatment and control. Under this definition of treatment and control, in some specifications, I can take advantage of using county-level data by controlling for state-by-year trends.⁷ A major advantage of county-level data over census data is that county-level data are available for intercensal years. Marriage rates, for example, are available for the years 1948, 1950, 1954, and 1960 at the county level while fertility is available at the yearly level from 1945–65. This allows me to control accurately for trends prior to the law change.

The disadvantage of using county-level data is that the county-level data do not contain details on sex, race, or age. In particular, data by age would have allowed for

7. Again, doing so would exclude Mississippi counties since all Mississippi counties are defined as a “treated” county.

sharper analysis because the minimum age and parental consent laws were directed toward younger age groups. In sum, the county-level analysis yields the preferred set of estimates. While using data by age and gender is an important check on the validity of the estimates, the major advantages of using county-level data include using distance from the Mississippi border, as well as data from intercensal years, and being able to control for state-specific trends..

The difference-in-differences strategy using county-level data can be estimated as follows:

$$(1) \text{ Outcome}_{ijt} = \beta_1 \text{Treat}_{ijt} + \beta_2 \text{Post}_{ijt} + \beta_3 (\text{Treat}_{ijt} \text{X} \text{Post}_{ijt}) + \alpha_{jt} + \gamma X_{ijt} + \varepsilon_{ijt}$$

where Outcome_{ijt} stands for an outcome like marriage or enrollment rates in county I in state j at time t . Treat_{ijt} denotes whether the county is defined as a “treated” county. In the case of using distance from the Mississippi border, Treat_{ijt} is simply the inverse of distance to the Mississippi border. An alternative definition of treatment used in this paper is to define all counties in Mississippi and border counties of neighboring states as treated. In this case, Treat_{ijt} is simply a dummy variable, taking the value of 1 for treated counties and 0 for control counties. Post_{ijt} is a dummy variable that takes a value of 1 for years after 1957 and is 0 otherwise. The coefficient of interest here is three, which is the difference-in-differences coefficient. X_{ijt} denotes a vector of controls like tractor use, employment in manufacturing, etc. α_{jt} denotes the state-by-year fixed effect, although I can only use this in cases where Mississippi is not included in the analysis. In the specifications that include Mississippi, I use state and year fixed effects.⁸

Due to the inclusion of border counties in neighboring states in the treatment group, standard errors are clustered at the county level. However, results using different levels of clusters are shown in the appendix. Bertrand, Duflo, and Mullainathan (2004) stress the importance of clustering standard errors while using DD techniques to examine the impact of law changes. Because the law was imposed at the state level in Mississippi, an argument could be made that conservative standard errors are achieved by clustering at the state level. However, clustering at the state level yields five clusters in my case leading to large standard errors for some of the results. To deal with a small number of clusters, I use three approaches. First, I use the methodology in Buchmueller, DiNardo, and Valleta (2011) of using “placebo” states to obtain the sampling distribution of the difference-in-differences coefficient. In other words, Equation 1 is estimated using other states as the treated state and the coefficient obtained for Mississippi is compared to the coefficient obtained for other states. According to Buchmueller, DiNardo, and Valleta (2011), this procedure results in “conservative and appropriate” standard errors. Second, I can increase the number of clusters under the standard procedure by expanding the control group to counties in Texas, Florida, Oklahoma, Virginia, West Virginia, Georgia, Kentucky, North Carolina, Delaware, and Maryland/Washington D.C. This gives me ten more states to include, increasing the number of clusters to 15. I could also include all states in the country and increase the number of clusters to 50. In general, different clustering procedures do not affect the interpretation of the main results. Even under the most conservative of clustering

8. This specification is:

$$\text{Outcome}_{ijt} = \beta_1 \text{Treat}_{ijt} + \beta_2 \text{Post}_{ijt} + \beta_3 (\text{Treat}_{ijt} \text{X} \text{Post}_{ijt}) + \alpha 1_t + \alpha 2_j + \gamma X_{ijt} + \varepsilon_{ijt}$$

procedures, most of the results remain statistically significant. Results with different clustering methods are presented in Appendix Table 2.

County-level data for birth rates and enrollment were obtained from the City and County Handbooks for the years 1948, 1950, 1954, and 1960. Data are not available for the intervening years. These data are at the county level and contain important demographic (marriage, fertility, and schooling) and labor market-related variables (wages, tractor use, etc.). The City and County Handbooks get their birth and marriage data from the Vital Statistics. However, not all variables are present for all years, nor are they necessarily in a format that is comparable across years. For example, school enrollment in the City and County Handbook is available for age groups 14–17 in 1950 but only for age groups five to 34 in 1960. Hence, school enrollment data at a comparable level between 1950 and 1960 are obtained from the historical census collection from the University of Virginia Library website. The historical census provides county identifiers, but as it is a census, there are no data on enrollment for any intercensal year.⁹ In addition, I obtained yearly county-level births from 1946–65 thanks to a data collection effort undertaken by Martha Bailey.¹⁰

B. State-Level Analysis

Data at the state level come from the census and have the advantage that some of these data contain data by age. Since the law change was intended to affect certain age groups more than others, using age-specific information, I can assign treatment status not only by state of residence but also by age. I can do this using census data from 1950 and 1960. Women below the age of 21 (and particularly below the age of 18) in 1957–58 were impacted by the change in marriage law, so women above the age of 21 in 1957 should not be affected by the change in marriage law, or at the very least should be affected less than women below the age of 21. This is because, while proof of age and blood test requirements affected everybody, the age restrictions were an added barrier for women below the age of 21 in 1960. In addition, verifying the results using census data is critical in light of potential misreporting of age in marriage licenses (Blank, Charles, and Sallee 2009). A triple difference-in-differences (essentially a state-cohort analysis) exploits the age-specific impact of the marriage law.

Using 1950 and 1960 census data at the state level, the estimating equations typically take the following form:

$$(2) \quad Y_{ijt} = \beta_g (Treat_{ijt} * Post_{ijt} * \sum_{g=33}^{g=14} A_{ijt}) + \gamma_1 (Treat_{ijt} * \sum_{g=33}^{g=14} A_{ijt}) + \gamma_2 (Post_{ijt} * \sum_{g=33}^{g=14} A_{ijt}) + \gamma_3 Treat_{ijt} + \gamma_4 Post_{ijt} + \gamma_5 \sum_{g=33}^{g=14} A_{ijt} + v_{ijt}.$$

Where Y_{ijt} is the relevant outcome (age of marriage, school enrollment, number of children, etc.) for person i in state j at time t belonging to age group g . $Treat$ is a dummy that takes on 1 if the person lives in Mississippi or its neighboring states

9. The historical censuses also do not have marriages and births by sex, race, or age. They simply contain aggregates for the years 1950 and 1960.

10. Bailey, Martha J. 2010. 1946–65 County Natality Data for AL, AR, MS, LA, TN. University of Michigan, December 31. Data collection funded by the University of Michigan's National Poverty Center, Robert Wood Johnson Health and Society Programs, University of Michigan Population Research Center's Eva Mueller Award, and the National Institute of Health (Grant HD058065-01A1).

and 0 otherwise.¹¹ *Post* is a dummy that is 1 if the year is 1960 and 0 otherwise. *A* is a dummy that takes on the value of 1 if the person belongs to age group *g* and 0 otherwise. The DD estimate is the triple interaction of *Age*, *Treat*, and *Post*, and all lower interaction terms and main effects are included in the regression. Note that the interaction of *Treat* and *Post* is not included. The inclusion of all age dummies interacted with *Treat* and *Post* implies that the double interaction is completely described by this triple interaction.

Triple DD estimates are obtained by comparing the β for younger versus older age groups. Standard errors are clustered at the state-year level. The age groups (as of 1957) are 11–14, 16–20, 21–25, and 26–30. These groupings were made to facilitate presentation of the results and to clearly show that the main impacts of the marriage law are coming through younger age groups.¹² Although the triple interaction for all age groups is included, double interactions and main effects for age group 26–30 form the excluded group. A key advantage of the census data is that in 1960 the census asked about age at first marriage for the sample that reports having been married. (In 1950, while they do not ask about age at first marriage, they ask about duration of marriage, from which we can impute the age of marriage for those who report currently being married.) Age of marriage in the census is likely to be more reliable than age of marriage in the Vital Statistics because the Vital Statistics records age of marriage at the time of marriage when incentives to misreport age might be higher. In the census, there are no such incentives to misreport.

The disadvantage of using census-level data is that there are no data for the intercensal years. This limits the extent to which I can control for differential trends in the 1950s. Moreover, it is difficult to accurately assign treatment status with just state-level identifiers. Because border counties of neighboring states of Mississippi were also affected, I present results using two versions of a treatment group. My preferred estimate comes from using just Mississippi and its neighboring states as the treatment group and the remaining states in the southern region (as defined by the census) as the control group. As a robustness check, I assign only Mississippi as the treatment group and treat its neighbors as the control group.

As a compromise between the county-level data and the census data, I collected data from the Vital Statistics at the state level for outcomes such as births and illegitimate births. The state-level Vital Statistics were available from 1952 onward and provide yearly information on births and illegitimate births by age and race of the mother. Although I cannot conduct a distance-level analysis with this data, I can estimate Equation 1 using Mississippi as the treated state and its neighbors as the control states for various age groups.

11. State of residence from the census is used to assign persons to treatment or control group. Because the law change was implemented in 1958, only two years passed between the passage of the law and the 1960 census. In 1960, 9 percent reported having lived in a different state in the last five years in the treatment group and 12 percent in the control group. Moreover, the migration rates are much lower for blacks than for whites. Only 3 percent of blacks in the treatment group in 1960 report having lived in a different state in the past five years while the same statistic for the control group is around 5 percent. The 1950 census does not have comparable migration information.

12. The groupings per se are not essential to the results; indeed, Appendix Table 3 shows very similar results when the age group dummies in Equation 2 are replaced with individual age dummies.

Table 2
Impact of Marriage Law and Distance from Mississippi Border

	Marriages 1	Log CBR 2	Enrollment 3
Distance inverse X post	138.37 [67.162]**	-11.19*** (1.587)	185.8*** (42.53)
Inverse of distance to Mississippi border	-240.53 [68.91]***	7.198*** (1.367)	-79.61** (39.34)
Post (=1 if year ≥ 1957)	-3.485 [1.645]**	-0.233*** (0.0238)	4.698*** (0.563)
Fixed effects used	State, Year		
Controls	Manufacturing wage, farms per 1,000 people, percent employed in manufacturing		
Mean of dependent variable	10.02	3.006	80.29
R-squared	0.08	0.505	0.427
Observations	903	903	586

Notes: Marriages are marriages per 1,000 in population, births are log births per 1,000 in population, and enrollment is the percentage of 14–17-year-olds enrolled in school. Distance (in miles) is computed using latitude-longitude information on counties and distance to the closest Mississippi county is computed using the Haversine formula. Mississippi is excluded from these regressions, as all points within Mississippi have a distance of 0. Counties from neighboring states of Mississippi used.

* significant at 10 percent; ** significant at 5 percent; *** significant at 1 percent, robust standard errors, clustered at the county level

IV. Results

This section presents results from estimating Equations 1 and 2. Because county-level estimates are the preferred estimates, for each outcome I first present county-level estimates followed by state-level estimates.

A. Marriage Decline

While Figures 1–4 largely establish the impact of the law change on marriage rates, this section provides more specific evidence. Using distance to the Mississippi border as the definition of treatment, Table 2 estimates Equation 1 and shows that distance to border matters for marriage rates after 1957 (the interaction of distance inverse and post dummy) and that this is a statistically significant effect. In order to make the table interpretable, I use the inverse of distance to Mississippi border; therefore, the interaction term suggests that places close to the Mississippi border experienced an increase in marriages after the law change. This confirms the casual observation in Plateris (1966, replicated in Appendix Table 1) as well as Figure 5. Note that this increase does not make up for the overall decrease in number of marriages.

The downside to this definition of treatment is that it necessarily excludes Mississippi from the analysis. In order to include Mississippi as a treated group, I define Mis-

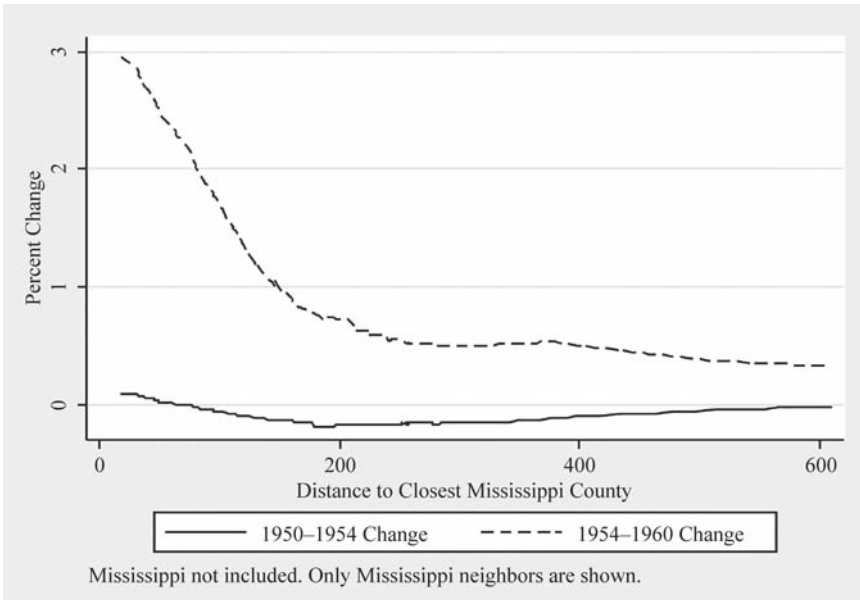


Figure 5
Percent Change in Number of Marriages

Notes: Data are from the County and City Data Book for the year 1950, 1954, and 1960. Distance is calculated using latitude and longitude coordinates of the counties, and the shortest distance to a Mississippi county is computed. Distance was computed using the Haversine formula and is in miles. Mississippi is excluded from this graph, while Louisiana, Arkansas, Tennessee, and Alabama are included.

Mississippi and bordering counties as the treatment group with remaining counties in the neighboring states as the control group. Table 3 shows estimates of Equation 1 with various controls. Although the treatment group definition stays the same (Mississippi and counties in Alabama, Arkansas, Tennessee, and Louisiana that border it), Column 1 uses all southern counties as the control group. Column 2 shows that restricting the control group to counties in the neighboring states of Mississippi does not change the results much. In Column 3, I add state and year controls, which again do not make any difference (I can use state fixed effects, as the treatment group consists of counties within states). Columns 4 and 5 add county-level controls like manufacturing wages, employment in manufacturing industries, number of farms, and employment in agriculture—these are key labor market-related variables, and trends in these variables could affect marriage rates independently of the change in law. Since agricultural employment was only collected for 1950 and 1960, Column 5 has fewer observations than Column 4. Comparing coefficients across Columns 1–5 shows that adding controls does not change the coefficient on the difference-in-differences estimator (the interaction of Treatment and Post dummies). The results indicate that after the change in law, treatment counties experienced a drop of ten marriages per 1,000 in the population. Prior to 1957, the rate of marriages per 1,000 in Mississippi and its

Table 3
Impact of Marriage Law on Marriages

	Marriages Per 1,000 in Population						
	All Treatment Counties Used				Mississippi Not Included		
	1	2	3	4	5	6	7
Post X treatment	-12.678 [3.656]***	-12.509 [3.749]***	-12.509 [3.756]***	-13.389 [4.027]***	-10.414 [4.077]**	4.96 [1.241]***	4.303 [0.910]***
Treatment (=1 if Mississippi or county bordering Mississippi)	11.657 [4.010]***	11.626 [4.104]***	-1.232 [1.319]	0.628 [1.501]	1.83 [2.509]	-5.6 [0.808]***	-5.309 [0.737]***
Post (=1 if year ≥ 1957)	-0.135 [0.451]	-0.305 [0.919]	-0.026 [1.093]	-2.188 [1.451]	-4.195 [1.748]**	-2.662 [1.413]*	2.902 [0.790]***
Manufacturing wage				-0.051 [0.039]	-0.042 [0.032]	-0.037 [0.032]	-0.051 [0.033]
Farms per 1,000 in population				1.186 [21.637]	20.608 [29.631]	11.794 [17.796]	1.978 [17.846]
Percent employed in manufacturing				0.047 [0.040]	0.005 [0.029]	0.048 [0.043]	0.044 [0.043]
Percent employed in agriculture					-0.037 [0.034]		

Fixed effects used Control group	None All counties in southern U.S.A.	None	State, Year		State, Year		State, Year	
			Counties in states neighboring Mississippi only		State, Year		State, Year	
Mean of dependent variable	12.04	13.05	12.82		12.98		10.05	
R-squared	0.01	0.04	0.09		0.08		0.08	
Observations	5,807	1,532	1,532		766		903	

Notes: Columns 1–5 use a state-year cluster, and Columns 6 and 7 use a treatment group, state-year cluster since Mississippi is excluded in these regressions. Dependent variable is marriages per 1,000 of the population. Data from city and county data book for years 1948, 1950, 1954, and 1960 are used. Population at the county level for 1950 and 1960 is used to compute the percentages (1954 numbers are divided by the population in 1950). Treatment is 1 if state is Mississippi and border counties. Percent employed in manufacturing and agriculture also uses population from 1950 and 1960 as the base. Agricultural employment data are only available for 1950 and 1960. Manufacturing employment and wages data are not available for 1948. Regression follows Equation 1 in the paper.

* significant at 10 percent; ** significant at 5 percent; *** significant at 1 percent, robust standard errors, clustered at the county level

neighboring counties was around 22 (Table 1). Table 2 shows a remarkable decline of almost 50 percent compared to the prelaw change average.

Columns 6 and 7 show the impact of the marriage law when we exclude Mississippi from the analysis. Because the entire state of Mississippi is part of the treatment group, when I include state-by-year fixed effects, only the border counties in the neighboring states are identified. Hence, the DD estimator has a positive sign for these columns. Comparing Columns 6 and 7 also shows that adding state-by-year fixed effects, which control for state-specific trends, does not change the fact that marriage rates were affected by the change in law. This suggests that the DD estimator is indeed picking up the change in marriages due to the change in law as opposed to changes due to other differential trends at the state level.

Moving to state-level analysis using the data from the census, Table 4 shows that for younger age groups among women, the probability of being married (this is defined throughout this analysis as being married by the time of the relevant census) decreased and the age of marriage increased by 1960. This table is created by estimating Equation 2 using five-year age groups instead of individual age dummies for easy interpretation. The omitted age group is the age group 31–35. Compared to this age group, Column 6 in Table 2 shows, for example, that black women in the age group 16–20 as of 1957 experienced an increase in marriage age of nearly 0.4 years. Compared to the average age of marriage for black women in the sample (19 years), the increase in marriage age is quite large. The same applies for the probability of being married. Black women in the age range 16–20 are thirteen percentage points less likely to report being married by 1960, representing a 23 percent change from the mean probability of being married.

Comparing the difference-in-differences coefficient across various age groups, it is clear that the most impacted age group is the group aged 16–20 in 1957–58 rather than, say, the age group of 26–30. This is reassuring as the changes in the marriage law were mainly directed at younger age groups due to various age restrictions.

Results using census data show that most of the overall effects appear to be driven by blacks. Blacks formed a large portion of the overall population in the South—by 1960, blacks were about 26 percent of the population in Mississippi and its surrounding states (Mississippi's population at the time was nearly 51 percent black). Moreover, among black women, the effect on marriages extends to the 21–25 age group as well. This is likely due to restrictions like proof of age and blood tests being a greater burden on blacks than on whites during this time period. This is similar to Buckles, Guldi, and Price (2009), who find that blood test laws for marriages were more of a deterrent to blacks. Thus, relative to whites, it appears that the marriage law changes in Mississippi raised the cost of marriage mainly for blacks.

B. Fertility

Figure 6 provides some visual evidence that, while there was no relationship between the change in births 1950–54 and distance to the Mississippi border, the largest drop in number of births per 1,000 in the population occurs near the border to Mississippi by 1960. As distance from the border increases, the drop in births appears to decrease. Column 2 in Table 2 shows that counties closer to the border after 1957 had a statistically significant drop in crude birth rates. (The estimation follows Equation 1 using the

Table 4
Impact of Marriage Law Using Census Data

Difference-in-Difference Coefficient, Age in 1957	Blacks				Whites			
	Probability of Marriage		Age of Marriage		Probability of Marriage		Age of Marriage	
	Female 1	Male 2	Female 3	Male 4	Female 5	Female 6	Female 7	Female 8
11–15	–0.013 [0.014]	–0.007 [0.003]*	–0.027 [0.092]	–0.21 [0.273]	–0.017 [0.024]	–0.126 [0.234]	–0.006 [0.014]	–0.007 [0.111]
16–20	–0.057 [0.023]**	–0.042 [0.023]*	0.154 [0.101]	–0.103 [0.107]	–0.139 [0.043]***	0.39 [0.159]**	–0.021 [0.025]	0.09 [0.123]
21–25	–0.014 [0.015]	–0.015 [0.014]	–0.069 [0.152]	0.061 [0.144]	–0.073 [0.035]**	–0.195 [0.491]	0.009 [0.015]	–0.013 [0.242]
26–30	–0.011 [0.010]	0.013 [0.013]	0.094 [0.285]	–0.055 [0.178]	–0.009 [0.023]	–0.245 [0.451]	–0.007 [0.010]	0.311 [0.320]
Mean of dependent variable	0.614	0.457	18.99	21.29	0.551	19.13	0.633	18.95
R-squared	0.46	0.48	0.17	0.21	0.42	0.18	0.47	0.17
Observations	175,045	168,111	58,763	43,851	39,867	10,302	134,610	48,265

Notes: Data from IPUMS 1 percent sample from years 1950 and 1960. All regressions are weighted by census person weights. Probability of marriage is a variable that is 1 if married, 0 otherwise at the time of the Census. Age of marriage is conditional on marriage. Regressions reflect estimation of Equation 2 in the paper. Controls included are all main effects of age groups and treatment group status.

* significant at 10 percent; ** significant at 5 percent; *** significant at 1 percent, robust standard errors, clustered at the state-year level

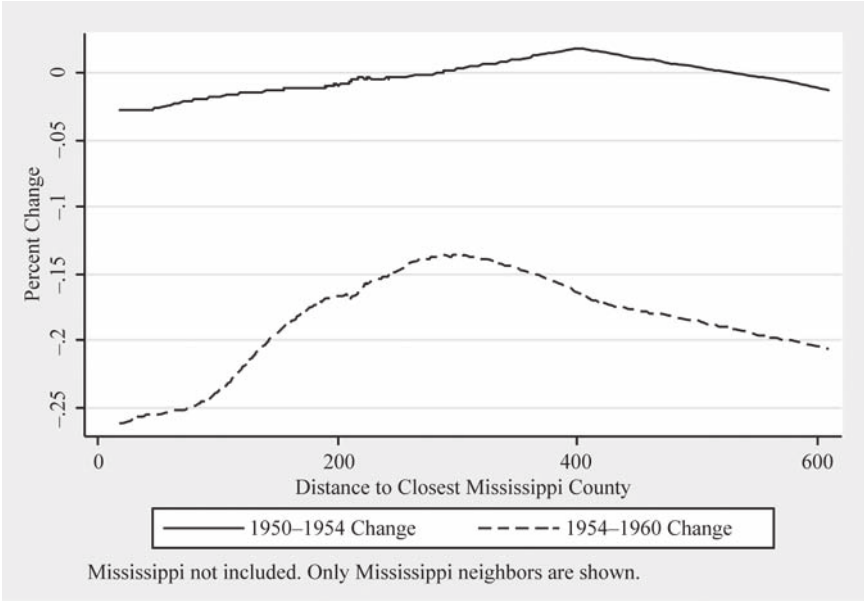


Figure 6
Percent Change in Number of Live Births

Notes: Data are from the County and City Data Book for the year 1950, 1954, and 1960. Distance is calculated using latitude and longitude coordinates of the counties, and the shortest distance to a Mississippi county is computed. Distance was computed using the Haversine formula and is in miles. Mississippi is excluded from this graph, while Louisiana, Arkansas, Tennessee, and Alabama are included.

log crude birth rate as the dependent variable.) However, unlike the graphs and figures used to show the decline in marriages, the drop in births is smaller in magnitude and, hence, is better represented in regression tables. Figure 6 shows that areas close to the border after the law change show greater negative changes (this was not the case before the law change) than areas further away.

Table 5 follows a similar estimation strategy as Table 3. Equation 1 is estimated using log of births per 1,000 in the population as the dependent variable. Once again, it is clear that adding controls does not change the coefficients. Although Table 3 shows that marriages increased slightly in the border counties, Table 4 shows that births decreased (Columns 6 and 7). This is not inconsistent at all, as the overall marriages in the border counties did decline. It is just that before the change in law, all the marriages were being recorded in Mississippi. The coefficients across the main specifications including Mississippi suggest a drop of around 5–6 percent. This drop in births is quite substantial considering that between 1910 and 1954 the drop in crude birth rate was around 16 percent for the entire country.

Tables 6a and 6b estimate Equation 1 using yearly county-level data. I choose to present this separately and not as my main specification as I do not have data for control variables like wages and employment at the yearly level. Table 6a uses these

yearly data and defines years after 1957 as years affected by the law change. The difference-in-differences coefficient in this table is negative throughout, even with the inclusion of state and state-by-year fixed effects. The results also seem robust to the inclusion of state linear and state nonlinear time trends (the coefficients of interest are just shy of significance at the 10 percent level in the first panel of Table 6a). The middle panel of 6a estimates the same relationships but restricts the control group to counties within 200 miles of the Mississippi border. These counties are arguably more similar to Mississippi (see Table 1) prior to the law change. Again, the results are largely consistent with the top panel. These results are not affected by the years chosen to be in the sample. In the bottom panel of this table, I restrict attention to the years 1955–60, and the effects are largely unchanged, suggesting that the effects are not driven by earlier or later time periods. An important way these results differ from the results in Table 5 is in the magnitude of the coefficients. The coefficients in Table 6a are quite a bit smaller than those in Table 5. One reason for this is the likely changes in 1954–60 not related to the marriage law change; hence, Table 5 might be playing up some of these larger reasons behind the fertility decline. The estimates in Table 6a suggest a decline in fertility of around 2 percent.

The other major advantage of the yearly data is that I can estimate a difference-in-differences estimate using each year from 1950–63 as a placebo year in which the law change occurred. In other words, I can treat each year as though it were a “treated” year and compute the difference-in-differences estimates. Hence, if events in 1956 Mississippi unrelated to the law change were driving the fertility results, the difference-in-differences coefficient for that year should be negative and significant. Table 6b shows that it is precisely in 1958 that the difference-in-differences coefficient becomes negative and statistically significant.¹³ In the years prior to 1958, the difference-in-differences coefficient is positive (although not statistically significant), indicating that the treated counties had a higher birth rate compared to the control counties, and this begins to change precisely in 1958. However, as is clear from the estimates, while consistent with the idea that the law change is driving these results, the existence of a mild pretrend (although not statistically significant) and the lack of a strict trend-break around 1957 in some specifications, should be treated with caution. Accounting for state-specific trends does not change the results much, except that some of the difference-in-differences coefficients are no longer statistically significant.

Another source of yearly data on fertility is the Vital Statistics tabulations of the number of births at the state level. One advantage of these tabulations over the yearly county-level data is that these tabulations are available by race and age. Tables 7a and 7b explore the fraction of births by race and age in a similar difference-in-differences setup as in Equation 1. Table 7a shows that the decline in births is among younger age groups and is larger in magnitude for blacks. Hence, it corroborates the results presented earlier showing that the law perhaps affected blacks more. The Vital Statistics also reports the number of illegitimate births by age and race for this time period. For younger women, I find an increase in illegitimate births after the passage of the law (Table 7b). Hence, while the law in effect prevented younger age groups

13. Note that each coefficient shown in this table comes from a separate regression.

Table 5
Impact of Marriage Law on Crude Birth Rate

	Log Births per 1,000 in Population						
	All Treatment Counties Used				Mississippi Not Included		
	1	2	3	4	5	6	7
Post X treatment	-0.0158 (0.0185)	-0.0424* (0.0216)	-0.0424* (0.0216)	-0.0584*** (0.0218)	-0.0558** (0.0274)	-0.136*** (0.0286)	-0.113*** (0.0307)
Treatment (=1 if Mississippi or county bordering Mississippi)	0.0781***	0.0903***	0.0487	0.0769**	0.0575*	0.104***	0.0952***
Post (=1 if year ≥ 1957)	(0.0213)	(0.0253)	(0.0397)	(0.0308)	(0.0324)	(0.0338)	(0.0358)
Manufacturing wage	-0.269*** (0.00609)	-0.242*** (0.0126)	-0.310*** (0.0144)	-0.320*** (0.0168)	-0.300*** (0.0185)	-0.322*** (0.0182)	-0.371*** (0.0221)
Farms per 1,000 in population				-0.00117** (0.000479)	-0.00139*** (0.000497)	-0.00142*** (0.000463)	-0.00130*** (0.000461)
				-3.222*** (0.228)	-4.138*** (0.408)	-3.336*** (0.277)	-3.353*** (0.280)

Percent employed in manufacturing Percent employed in agriculture	None		State, Year		State, Year		State, Year	
	All counties in southern U.S.A.	None	Counties in states neighboring Mississippi only		State, Year		State, Year	
Fixed effects used								
Control group								
Observations	4,650	1,372	1,372	1,029	686	810	810	810
R-squared	0.139	0.182	0.371	0.501	0.503	0.514	0.541	0.541

Notes: Dependent variable is log births per 1,000 of the population. Data from city and county data book for years 1948, 1950, 1954, and 1960 are used. Population at the county level for 1950 and 1960 is used to compute the percentages (1954 numbers are divided by the average population between 1950 and 1960). Treatment is 1 if state is Mississippi and border counties. Agricultural employment data are only available for 1950 and 1960. Manufacturing employment and wages data are not available for 1948. Regressions reflect estimation of Equation 1 in the paper. Outliers in the dependent variable (top and bottom 5 percent of the counties with changes in birth rates) removed. Including these counties increases the magnitude of the effect to around 8–9 percent in Columns 1–5 and the effects remaining statistically significant. These estimates are presented in Appendix Table 13.

* significant at 10 percent; ** significant at 5 percent; *** significant at 1 percent, robust standard errors, clustered at the county level

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Table 6a
Log Birth Rates Per 1,000 of Female Population

	1	2	3	4	5
Full sample 1950–63					
Difference-in-difference	–0.0179** (0.00850)	–0.0179** (0.00851)	–0.016 (0.01)	–0.0159 (0.0100)	–0.00828 (0.0135)
Dummy for post (year > 1957)	0.00643 (0.00529)	0.0218*** (0.00707)	0.0164** (0.00656)	0.0241*** (0.00595)	0.0108 (0.0135)
Dummy for treatment group	0.174*** (0.0200)	0.0924** (0.0361)	0.0916** (0.0365)	0.0916** (0.0365)	0.0883** (0.0374)
Other controls	None	State & Year FE	State linear time trends	State quadratic time trends	State X Year FE
Mean of dependent variable	4.791				4.744
Observations	5,362	5,362	5,362	5,362	4,214
R-squared	0.130	0.313	0.314	0.315	0.303
Full sample 1950–63, within 200 miles of Mississippi					
Difference-in-difference	–0.0240** (0.00938)	–0.0240** (0.00940)	–0.0204* (0.0114)	–0.0204* (0.0114)	–0.0183 (0.0149)
Dummy for post (year > 1957)	0.0126* (0.00660)	0.0401*** (0.00858)	0.0186** (0.00847)	0.0315*** (0.00794)	–0.0757*** (0.0185)
Dummy for treatment group	0.119*** (0.0229)	0.0709** (0.0353)	0.0694* (0.0357)	0.0693* (0.0357)	0.0684* (0.0365)

Other controls	None	State & Year FE	State linear time trends	State quadratic time trends	State X Year FE
Mean of dependent variable	4.848				4.814
Observations	3,556	3,556	3,556	3,556	2,408
R-squared	0.071	0.223	0.219	0.222	0.251
Restricted to 1955–60					
Difference-in-difference	–0.0218*** (0.00782)	–0.0218*** (0.00783)	–0.0102 (0.00987)	–0.0102 (0.00987)	–0.0195 (0.0124)
Dummy for post (year > 1957)	0.0149*** (0.00513)	0.0185*** (0.00684)	–0.0168** (0.00702)	–0.0168** (0.00702)	0.00424 (0.0102)
Dummy for treatment group	0.181*** (0.0219)	0.0924** (0.0389)	0.0866** (0.0399)	0.0866** (0.0399)	0.0913** (0.0408)
Other controls	None	State & Year FE	State linear time trends	State quadratic time trends	State X Year FE
Mean of dependent variable	4.795				4.759
Observations	2,298	2,298	2,298	2,298	1,806
R-squared	0.128	0.337	0.340	0.340	0.330

Notes: Data used are country level log fertility rates per 1,000 of the female population. Treatment group is all of Mississippi and counties in neighboring states sharing a border with Mississippi. Control groups are all other counties in neighboring states of Mississippi. Since all of Mississippi is considered part of the treatment group, regressions that use StateXYear fixed effects drop Mississippi and only use border counties as the treatment group. Regressions reflect estimation of Equation 1 in the paper. Robust standard errors in brackets.

* significant at 10 percent; ** significant at 5 percent; *** significant at 1 percent

Table 6b*Log Birth Rates Per 1,000 of Female Population*

Difference-in-Difference Coefficient	Full Sample		Sample Within 200 Miles of Mississippi Border	
	1	2	3	4
1951	0.000464 (0.0108)	0.00188 (0.0189)	0.0131 (0.0115)	0.00421 (0.0190)
1952	-0.00290 (0.00843)	-0.0251* (0.0141)	-0.000120 (0.00957)	-0.0184 (0.0152)
1953	0.0179** (0.00855)	0.0134 (0.0141)	0.0116 (0.00940)	0.0157 (0.0148)
1954	0.0140 (0.00849)	0.0117 (0.0135)	0.0118 (0.00869)	0.0128 (0.0142)
1955	0.0363*** (0.00806)	0.0251* (0.0133)	0.0414*** (0.00889)	0.0335** (0.0130)
1956	0.0148 (0.00918)	0.0108 (0.0161)	0.0170 (0.0107)	0.0242 (0.0159)
1957	-0.00641 (0.00820)	-0.0149 (0.0151)	-0.0136 (0.00926)	-0.0127 (0.0156)
1958	-0.0138* (0.00797)	-0.0148 (0.0132)	-0.0303*** (0.00817)	-0.0266** (0.0125)
1959	-0.000617 (0.00916)	-0.0207 (0.0189)	-0.00680 (0.0102)	-0.0274 (0.0201)
1960	-0.0114 (0.00954)	-0.00649 (0.0130)	-0.0167 (0.0103)	-0.0121 (0.0139)
1961	-0.0141 (0.00891)	-0.00436 (0.0149)	-0.0175* (0.0103)	-0.0124 (0.0158)
1962	-0.0123 (0.0113)	0.00987 (0.0167)	-0.0120 (0.0119)	0.00437 (0.0176)
1963	-0.0137 (0.0110)	0.00594 (0.0154)	-0.00534 (0.0121)	0.00654 (0.0164)
Controls	State, Year FE	StateXYear FE	State, Year FE	StateXYear FE
Observations	5,362	4,214	3,556	2,408

Notes: Reported coefficients are difference-in-difference coefficients for each year. Separate regressions are estimated for each coefficient in a specification similar to Equation 1. Treatment and control group definitions are the same as in Table 5. Columns including StateXYear FEs have fewer observations as Mississippi is omitted from those regressions (as *all* of Mississippi is treated as opposed to just border counties in neighboring states).

Robust standard errors in brackets.

* significant at 10 percent; ** significant at 5 percent; *** significant at 1 percent

Table 7a
Impact of Marriage Law on Births by Age of Mother

	Percent of Total Births by Age of Mother		
	Under 15 1	15–19 2	20–24 3
Nonwhites			
Difference-in-difference estimate (post X treatment)	–0.064 [0.046]	–0.765 [0.316]**	0.556 [0.334]
Treatment (=1 if Mississippi or neighbors, 0 otherwise)	–0.119 [0.024]***	–0.92 [0.150]***	–2.124 [0.169]***
Post (=1 if year > 1957)	0.066 [0.030]*	0.301 [0.336]	–0.683 [0.336]*
Mean of dependent variable			
R-squared	0.66	0.83	0.83
Observations	125	125	125
Whites			
Difference-in-difference estimate (post X treatment)	–0.023 [0.006]***	–0.153 [0.214]	0.772 [0.247]**
Treatment (=1 if Mississippi or neighbors, 0 otherwise)	–0.014 [0.003]***	–0.33 [0.115]**	1.39 [0.133]***
Post (=1 if year > 1957)	0.04 [0.014]**	2.331 [0.222]***	1.646 [0.219]***
Mean of dependent variable			
R-squared	0.68	0.96	0.94
Observations	129	129	129

Notes: Dependent variable is percent of total births that are born to mothers of the relevant age group. Data are from the Vital Statistics, for years 1952–64. Treatment group is Mississippi, Alabama, Louisiana, and Tennessee. Arkansas did not report statistics for this variable during this period. The control group consists of Florida, North Carolina, Kentucky, Virginia, West Virginia, and Texas. Georgia, Delaware, and the District of Columbia did not consistently report this variable for these years. Regressions reflect estimation of Equation 1 in the paper. Only controls used are state and year fixed effects.

* significant at 10 percent; ** significant at 5 percent; *** significant at 1 percent, robust standard errors, clustered at the state level

from marrying and having children, it did have some unintended consequences as it raised the number of illegitimate children.

Data from the census reveal that women affected by the law change in Mississippi were less likely to have children compared to women not affected by the law change. Table 8 shows that, compared to older groups in control counties, black women in the age group 16–20 are nearly ten percentage points less likely to have a child. The fraction of black women in 1950 in this age range that had children was around 43 percent. Hence, relative to that mean, this is a large effect. Comparing the DD coefficient of this age group to the age group 26–30 shows that it was the younger groups that were most affected. As in the case of marriage rates, blacks appear more affected than

Table 7b
Impact of Marriage Law on Illegitimate Births

	Percent Illegitimate (of Total) by Age of Mother		
	Under 15	15–19	20–24
Percent illegitimate — nonwhites			
Difference-in-difference estimate (post X treatment)	5.885 [2.974]*	1.277 [1.507]	0.65 [1.319]
Treatment (=1 if Mississippi or neighbors, 0 otherwise)	9.088 [1.669]***	4.585 [1.053]***	6.956 [1.132]***
Post (=1 if year > 1957)	–12.332 [1.455]***	0.952 [0.777]	–1.384 [0.704]*
Mean of dependent variable			
R-squared	0.42	0.94	0.73
Observations	128	127	128
Percent illegitimate — whites			
Difference-in-difference estimate (post X treatment)	–3.91 [6.763]	–0.423 [0.540]	–0.279 [0.149]*
Treatment (=1 if Mississippi or neighbors, 0 otherwise)	8.024 [5.178]	2.529 [0.611]***	1.247 [0.213]***
Post (=1 if year > 1957)	–5.74 [3.642]	–1.848 [0.291]***	–1.044 [0.080]***
R-squared	0.35	0.78	0.88
Observations	129	130	130

Notes: Dependent variable is percent of total births that are illegitimate to mothers of the relevant age group. Data are from U.S. Vital Statistics for years 1952–64. Treatment group is Mississippi, Alabama, Louisiana, and Tennessee. Arkansas did not report statistics for this variable during this period. The control group consists of Florida, North Carolina, Kentucky, Virginia, West Virginia, and Texas. Georgia, Delaware, and the District of Columbia did not consistently report this variable for these years. Regressions reflect estimation of Equation 1 in the paper. State and year fixed effects are the only controls.

* significant at 10 percent; ** significant at 5 percent; *** significant at 1 percent, robust standard errors, clustered at the state level

whites and seem to be driving the overall results. This is perhaps another reason why the results in the overall county-level data sets are perhaps a bit muted in comparison. County-level data by race might show sharper decreases.

Taken together, the results from county- and state-level data support the idea that increasing barriers to marriage led to a decrease in birth rates, at least in the immediate short run, by the early 1960s. Later in this section, I examine some of the long-run consequences on fertility due to the change in marriage law in Mississippi. However, given the larger declines in fertility that were taking place all over the country during this period, some of the declines using broader differences (1954–60, or 1950–60) could be capturing more than just the changes due to the law change. However, the results by distance to the Mississippi border, analysis by age of the mother, and the

Table 8
Impact of Marriage Law Using Census Data

	Overall			Blacks		Whites	
	Number of Children 1	Probability of One Child 2	Number of Children 3	Probability of One Child 4	Number of Children 5	Probability of One Child 6	
Difference-in-Difference Coefficient, Age in 1957							
11–15	–0.006 [0.012]	–0.004 [0.008]	–0.024 [0.022]	–0.016 [0.013]	0.007 [0.008]	0.006 [0.006]	
16–20	–0.088 [0.054]	–0.04 [0.025]	–0.215 [0.079]**	–0.091 [0.030]***	–0.044 [0.058]	–0.019 [0.032]	
21–25	–0.052 [0.101]	–0.028 [0.022]	–0.089 [0.170]	–0.083 [0.035]**	–0.068 [0.092]	–0.021 [0.022]	
26–30	–0.018 [0.140]	–0.006 [0.017]	0.366 [0.245]	0.037 [0.031]	–0.162 [0.113]	–0.025 [0.017]	
Mean of dependent variable	1.199	0.490	1.291	0.422	1.171	0.510	
R-squared	0.29	0.33	0.24	0.24	0.32	0.36	
Observations	175,045	175,045	39,867	39,867	134,610	134,610	

Notes: Data from IPUMS 1 percent sample from years 1950 and 1960. All regressions are weighted by census person weights. Number of children is the variable “nchild” from the census. Using “chborn” does not change the results. Regressions reflect estimation of Equation 2 in the paper. Controls included are all main effects of age groups and treatment group status.

* significant at 10 percent; ** significant at 5 percent; *** significant at 1 percent, robust standard errors, clustered at the state-year level

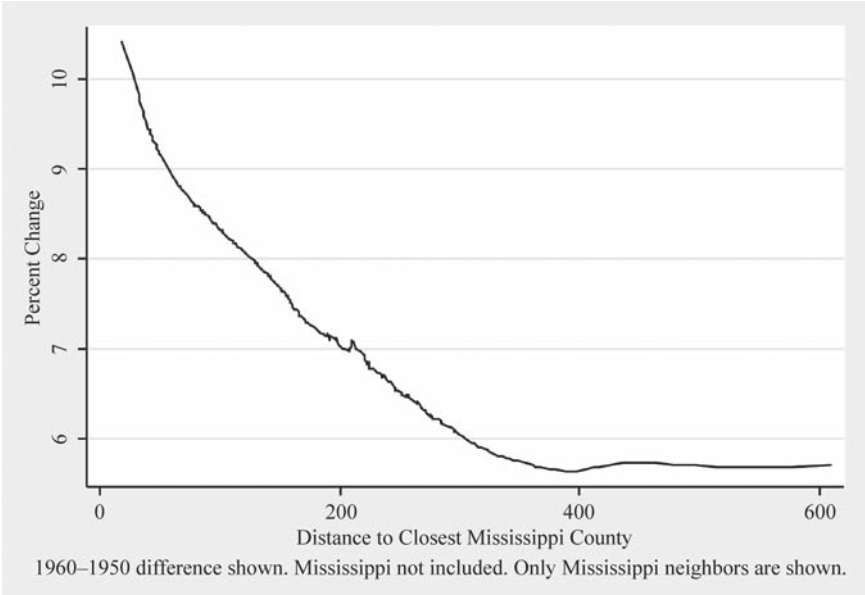


Figure 7
Difference in Percentage of 14–17-Year-Olds Enrolled in School

Notes: Data are from the Historical Census of 1950 and 1960 at the county level. Distance is calculated using latitude and longitude coordinates of the counties, and the shortest distance to a Mississippi county is computed. Distance was computed using the Haversine formula and is in miles. Mississippi is excluded from these graphs. Louisiana, Arkansas, Tennessee, and Alabama are included. Data for intercensal years are not available.

results using county-level yearly data should mitigate concerns that forces other than the marriage law are driving all the results.

C. School Enrollment

Goldin and Katz (2002) find that when women have control over their fertility, they invest more in their career. Does a change in marriage law discouraging early marriage have a similar effect? According to Field and Ambrus (2006), a mandated increase in the minimum age of marriage should result in greater educational attainment for women. Using the same empirical strategies as before, I examine whether the change in law led to greater educational attainment and enrollment. Unfortunately, data on school enrollment are not as extensive for this period as the data on fertility. For example, there are no comparable yearly county- or state-level data on enrollment rates for all treatment and control areas.

The county-level data for enrollment only exist for 1950 and 1960 (this is because comparable enrollment data were only available via the historical census). Hence, I cannot produce a graph similar to Figures 5 and 6; instead, in Figure 7, I plot the 1960–50 difference against distance to the Mississippi border. The graph indicates a

strong trend similar to Figure 5, in that areas close to the border experience the highest gains in enrollment. Column 3 in Table 2 shows that counties closer to the border after 1957 had statistically higher enrollment rate gains compared to counties further away (the estimations follow Equation 1 using enrollment rates as the dependent variable).

Table 9 shows estimates of the DD coefficients for school enrollment. Similar to Tables 3 and 5, I estimate Equation 1 using percentage of 14–17-year-olds enrolled in school as the dependent variable. The DD estimates across all specifications are robust to the addition of controls and suggest that the change in marriage law led to a 2.5 percentage point increase in school enrollment for this age group. Because overall school enrollment rates were quite high, this represents a small increase of 3 percent over the mean enrollment at that time. During the decade of 1950–60, there were no changes to compulsory schooling laws in Mississippi or its neighboring states (Dahl 2009). These findings are in line with Field and Ambrus (2006), who posit that a compulsory increase in marriage age will lead to greater school attainment. Unfortunately, the county-level data do not permit an analysis of schooling attainment by 1960 because schooling attainment data are only collected for people who are too old to be affected by the law.

Using data from the census, I show that black enrollment rates for the younger age groups were the most impacted. Compared to older age groups and to the control states, Table 10 shows that black men and women between ages 16–20 had higher enrollment rates in 1960. As in the case of the other outcomes already examined, the overall results appear to be driven by the results for blacks. A concern might be that these results are driven by changes in the quality of education during this period in treatment compared to control groups. In Appendix Figures 1 and 2, I show that treatment groups did not have a differential change in conventional school quality measures during this time period. Moreover, Appendix Table 7a shows that employment and wages did not change differentially in treatment areas during this time period, thus making it unlikely that changes in returns to education explain the increase in enrollment rates. (I explain Appendix Table 7a in detail in the section on robustness checks.)

D. Evidence From Law Changes in Other States

In 1957, a few other states (not those in our treatment or control group) also instituted changes in their marriage laws. These changes, however, did not alter the minimum age of marriage but instead increased premarital requirements as in the case of South Carolina, where a parental affidavit was required for parties younger than 16. Other states more broadly instituted requirements for blood tests (some of these were repealed in the 1980s as discussed in Buckles, Guldi, and Price 2011). The states that experienced a change in law around the same time period were Arizona, New Mexico, Iowa, Indiana, and South Carolina (Plateris 1966).¹⁴ Using the same strategy as in

14. It is difficult to pin down precisely what the changes in law in these other states were. Examining law citations from WestLaw and LexusNexus, it appears that the change in law in Arizona, New Mexico, and Indiana was a change in blood test requirement while the change in South Carolina was a change in parental consent. I was not able to find a similar citation for the Iowa law change.

Table 9
Impact of Marriage Law on School Enrollment

	Percent of 14–17-Year-Olds Enrolled in School						
	All Treatment Counties Used						Mississippi Not Included
	1	2	3	4	5	6	7
Post X treatment	2.65 [0.501]***	2.658 [0.573]***	2.658 [0.575]***	2.547 [0.599]***	2.244 [0.615]***	2.857 [1.121]**	2.075 [0.943]**
Treatment (=1 if Mississippi or county bordering Mississippi)	2.029 [0.549]***	0.745 [0.620]	-1.608 [0.858]*	-1.34 [0.801]*	-0.656 [0.755]	-1.485 [0.977]	-1.073 [0.933]
Post (=1 if year ≥ 1957)	6.725 [0.225]***	6.694 [0.357]***	6.694 [0.358]***	5.985 [0.430]***	4.735 [0.450]***	5.941 [0.457]***	6.825 [0.668]***
Manufacturing wage				0.027 [0.013]**	0.03 [0.014]**	0.03 [0.014]**	0.03 [0.015]**
Farms per 1,000 in population				-10.799 [7.630]	30.404 [11.814]**	-11.361 [9.276]	-7.806 [9.340]
Percent employed in manufacturing				0.024 [0.008]**	0.011 [0.007]	0.026 [0.009]**	0.031 [0.009]**
Percent employed in agriculture					-0.049 [0.010]***		

Fixed effects used Control group	None All counties in southern U.S.A.	None	State, Year			State, Year			State X Year
			Counties in states neighboring Mississippi only			Mississippi only			
Mean of dependent variable	79.04	80.85				80.3			
R-squared	0.2	0.31	0.42	0.45	0.48	0.42			0.45
Observations	2,476	750	750	750	750	586			586

Notes: Dependent variable is percent of 14–17-year-olds who are currently enrolled in school. Data from the historical census of 1950 and 1960 are used. Treatment is 1 if state is Mississippi and border counties. Percent employed in manufacturing and agriculture uses population from 1950 and 1960 as the base. Agricultural employment data are only available for 1950 and 1960. Regressions reflect estimation of Equation 1 in the paper.
 * significant at 10 percent; ** significant at 5 percent; *** significant at 1 percent, robust standard errors, clustered at the county level

Table 10
Impact of Marriage Law Using Census Data

	Overall				Blacks		Whites	
	Enrollment		Highest Grade Completed		Enrollment	Highest Grade Completed	Enrollment	Highest Grade Completed
	Female 1	Male 2	Female 3	Male 4				
Difference-in-Difference Coefficient, Age in 1957					Female 5	Female 6	Female 7	Female 8
11–15	0.022 [0.023]	–0.01 [0.029]	0.219 [0.166]	0.242 [0.171]	0.019 [0.048]	–0.116 [0.290]	0.02 [0.026]	0.274 [0.165]
16–20	0.043 [0.015]***	–0.004 [0.026]	0.388 [0.184]**	0.327 [0.344]	0.097 [0.037]**	0.053 [0.360]	0.024 [0.021]	0.397 [0.255]
21–25	0.012 [0.010]	0.019 [0.013]	–0.027 [0.194]	0.248 [0.268]	0.011 [0.016]	–0.004 [0.495]	0.011 [0.013]	0.006 [0.248]
26–30	–0.008 [0.016]	–0.013 [0.033]	–0.093 [0.194]	0.008 [0.223]	0.005 [0.031]	–0.499 [0.521]	–0.013 [0.016]	0.14 [0.274]
Mean of dependent variable	0.271	0.326	7.176	7.119	0.287	5.785	0.266	7.589
R-squared	0.49	0.39	0.49	0.46	0.46	0.53	0.5	0.48
Observations	92,666	90,218	175,045	168,111	18,552	39,867	73,796	134,610

Notes: Data from IPUMS 1 percent sample from years 1950 and 1960. All regressions are weighted by census person weights. Regressions reflect estimation of Equation 2 in the paper. Controls included are all main effects of age groups and treatment group status.
* significant at 10 percent; ** significant at 5 percent; *** significant at 1 percent, robust standard errors, clustered at the state-year level

Equation 2 and using census data, we can examine whether these changes in law resulted in decreases in marriage rates and fertility.

Table 11 suggests that marriage laws had little impact in Arizona, New Mexico (I combine these states into one treatment group since they share a border), and Indiana on the probability of marriage for different age groups. However, in Iowa there does appear to be a sizable effect concentrated in the early age groups, just as in Mississippi. However, Mississippi had an equally sizable negative effect for the next age category as well (this coefficient is just shy of significance at the 10 percent level). Note that the results for Mississippi are different in this table as I use a more restricted version of treatment and control (Mississippi is treatment and neighboring states are control). This is to make it compatible with the other states examined, where I use just the state as the treated area and its neighbors as the control. Comparing the results of other states to that of Mississippi shows that the law changes in Mississippi had a much bigger impact than elsewhere. This is likely due to the age and other restrictions included in the new Mississippi law; these were not features of the law changes in other states, which focused mainly on blood tests. It is also likely that, because blacks were more impacted by the law change (blood tests and other requirements were likely more expensive for blacks) and blacks comprised a significant fraction of the population in Mississippi, we see larger effects in Mississippi.

E. Long-Run Outcomes

Using the census of 1970 and 1990, I examine the long-run impacts of the law change. I do this by using age in 1957 as a variable that defines treatment, along with residence in Mississippi and neighboring states. For example, I consider a 17-year-old living in Mississippi in 1957 as treated and a 30-year-old in Mississippi in 1957 as untreated. The difference-in-differences comes from using other states in the southern United States as my control states. Moreover, I examine all long-run outcomes only for black females since they are the most affected group. The short-run effects for whites are quite muted, so the long effects are equally negligible.

Appendix Table 5 shows that people affected by the law in the relevant states had, by 1970, a higher age at marriage and fewer children. However, there is no impact on the probability of marriage or the probability of having a child. Hence, these results are very consistent with the notion that marriage laws affect the timing of marriage and fertility but not the extensive margin of these decisions themselves. There appears to be a positive effect on high school completion although this effect is not significant. Panel B of Appendix Table 5 conducts a placebo test to ensure that the long-run results are not picking some inherent trend in reporting by 1970. Using age groups that were not affected by law, I find no evidence of similar results for the placebo group. Indeed, the few results that are significant go in the opposite direction.

In Appendix Table 6, I use the 1990 census to examine whether the results seen in 1970 still persist. This table reveals that by 1990 the difference in number of children born disappears, suggesting that the fertility effects seen in earlier censuses are largely the effects of timing. As one might expect, however, the education results appear to be persistent. The analyses of the 1970 and 1990 census suggest that the Mississippi marriage law change largely had an impact of delaying marriage and fertility. However, it does appear to have had a lasting impact on years of schooling.

Table 11
Impact of Marriage Law Using Census Data

Difference-in-Difference Coefficient, Age in 1957	Probability of Marriage: Women				
	Mississippi 1	Arizona and New Mexico 2	Indiana 3	South Carolina 4	Iowa 5
11–15	–0.0362** (0.0179)	0.00273 (0.0202)	–0.0106 (0.0118)	0.00733 (0.0163)	–0.0249* (0.0136)
16–20	–0.0395 (0.0262)	–0.0274 (0.0274)	–0.0161 (0.0199)	–0.0269 (0.0267)	–0.0121 (0.0261)
21–25	–0.0189 (0.0184)	0.0136 (0.0189)	0.0330*** (0.0122)	0.00647 (0.0184)	–0.0190 (0.0160)
26–30	–0.00322 (0.0141)	–0.0208 (0.0151)	–0.0260*** (0.00937)	0.0131 (0.0164)	–0.0121 (0.0136)
Mean of dependent variable	0.604	0.649	0.611	0.591	0.600
Observations	49,517	83,779	104,196	39,553	65,760

Notes: Data from IPUMS 1 percent sample from years 1950 and 1960. Regressions represent estimation of Equation 2 in the paper. For each state in the columns, the neighboring states are used as a control group. Please see appendix for list of control states used for each treated state. Controls included are all main effects of age groups and treatment group status.

* significant at 10 percent; ** significant at 5 percent; *** significant at 1 percent, robust standard errors, clustered at the state-year level

V. Robustness Checks

If there are differential trends in treatment and control groups along any outcome that could be related to marriage, fertility, and schooling, then the results in the previous sections could be driven by those trends rather than the change in marriage law. In this section, I explore various labor market and technology-related outcomes for the treatment and control groups. In his analysis of black teenage employment in the South from 1950–70, Cogan (1981) posits that “technological progress is the principal cause of the agricultural employment decline among black youth.” Hence, we might worry that the treatment and control counties have differential trends in the adoption or use of various agricultural technologies, leading to differential trends in marriage, fertility, and schooling.

Appendix Table 7a shows a DD estimate for various labor market and technological outcomes. Manufacturing employment, manufacturing wage, and tractor use on farms do not seem to have changed differentially in treatment counties after the passage of the marriage law. However, the number of farms per 1,000 in the population as well as agricultural employment seem to have differentially decreased after the passage of the law. Including these variables directly in regressions in Tables 3, 5, and 9 does not change the DD estimate. Moreover, the percent of people employed in agriculture seems to have no statistically significant effect on marriages or the crude birth rate (Column 5 in Tables 3 and 5). The rate of farms per 1,000 in the population does not have a statistically significant effect on marriages (Column 6, Table 3). Moreover, including a state-by-year fixed effect seems to negate any effect farms per 1,000 might have on school enrollment (Column 7, Table 9). However, to examine this further, I add the interactions of farms per 1,000 with the *Post* and *Treat* dummy to the set of controls in Column 5 in Tables 3, 5, and 9. The idea behind this is that if farms per 1,000 before and after the change in law was the real mover in marriage, fertility, and enrollment, including the interaction of farms per 1,000, then the *Post* dummy should capture the entire treatment effect. Adding these controls makes no differences to the original DD estimates (results not reported, available upon request).

In Appendix Table 7b, I randomly assign treatment status to counties within the states of Louisiana, Arkansas, Mississippi, Tennessee, and Alabama. With a 1,000 repetitions of regressions of the form of Equation 1, I am able to reject that random assignment of treatment can generate the results obtained by the DD estimate.

A concern I was able to account for effectively in the county-level analysis was technological advancement in agriculture in the treatment versus the control group. States like Mississippi, its neighboring states, and Texas (one of the states in the control group) mechanized much more rapidly than other Southern states. It is important to control for cotton mechanization because the advent of mechanization could have impacted returns to schooling, which in turn might affect decisions to marry, bear children, and enroll in school. Moreover, cotton mechanization was highly correlated with migration during this period (Heinicke 1994), and this migration mainly involved blacks moving out of the South.

Appendix Table 8 shows that Arkansas, Louisiana, Mississippi, and Texas mechanized much faster than other Southern states. Among these, only Texas is in the control group. To account for cotton mechanization, I simply can redefine the control group as consisting only of Texas and omitting Tennessee from the treatment group. Table

9 in the appendix shows that for black women between the ages of 19–23 in the treatment group, marriage rates and fertility declined while educational enrollment and attainment increased compared to the (new) control group. Hence, accounting for potentially confounding factors like the effects of cotton mechanization does not significantly alter the earlier results even at the state level.

To show that the enrollment rate increases are not explained by differential school quality, I plot various school input data at the yearly level for the treatment and control groups. Figures 1 and 2 in the appendix show that there were no differential changes in school quality after 1957 in treatment and control groups.

A. Appropriateness of Control Groups

I can test the robustness of the results to alternative ways of creating control groups. In Appendix Figures 3 and 4, I show graphs of treatment versus control groups when I create synthetic control groups using the methodology of Abadie, Diamond, and Hainmueller (2010). Synthetic control groups are constructed using yearly county-level birth rate data (because such data at the yearly level are only available for birth rates), where birth rates before 1957 in the treatment group are used to find counties in the control group that most closely resemble the birth rate trends in the treatment group. Weights are then assigned in the control group, providing the greatest weight to counties that more resemble the treatment group's trends. Because there are many counties in the control group, I create units by distance to the Mississippi border. Hence, a distance of 0 is the treatment group and among a set of potential "distance" based control groups, weights are assigned. This procedure places more weight on counties close to the Mississippi border (as was expected based on the discussion above). The graphs suggest that, while the procedure matches quite well on prelaw change trends, the trends after the law change diverge, suggesting a drop in birth rates in treatment relative to control groups.

Another way to construct control groups is to use propensity matching techniques to place weights on counties in the control group that match the counties in the treatment group on observables. For this procedure, I use the data from the City and County Handbook to match on a large set of observables like marriage rates, birth rates, percent farms, percent employed in manufacturing and agriculture, and the manufacturing wage from 1950 and 1954. As Appendix Table 11 suggests, while birth and marriage rates are not statistically different pre-1960, they are statistically different in 1960 when weighted using the inverse propensity weights. Hence, even with a more flexible way of constructing control groups, the essence of the results is retained.

VI. Conclusions

This paper shows that raising the cost of marriage can have large impacts on marriage rates, crude birth rates, and school enrollment. Hence, barriers to marriage for women can result in delayed fertility, and if early marriage can be delayed, can even lead to higher school enrollment. The change in marriage law in Mississippi in 1957–58 involved an increase in the minimum age of marriage, parental consent requirements, and other restrictions like blood tests, proof of age, and a com-

pulsory waiting period. Most of the effects appear concentrated on young women as the law was aimed at preventing marriage and delaying childbearing at a young age. However, I do find a small increase in illegitimacy after the law was passed. School enrollment rates experienced a small increase as a result of the law. As Field and Ambrus (2006) suggest, an increase in minimum age could be one way of creating barriers that could be beneficial for women's educational outcomes.

However, in large part, the law change had bite in this context because the United States has good legal enforcement, unlike other settings, in particular developing countries where knowledge of such law changes is scarce. For example, in India, the minimum marriage age for women is 18, yet in a survey of nearly 90,000 married women conducted in 1992–93, only 35 percent were aware of the law. However, in such societies premarital sex and childbearing out of wedlock is rather taboo so postponing marriage would likely delay fertility and raise schooling. Hence, with better enforcement of laws, it is likely that even in the developing country context changes in marriage law would have similar effects to what I find in the case of Mississippi.

This paper contributes to the understanding of how public policy on marriages can affect important outcomes like fertility and education. While cohabitation rates and out-of-wedlock births have been on the rise, marriage still occupies a central role in American society. However, further research is needed to analyze why raising the cost of marriage (apart from minimum age laws) can alter the decision to marry. If people take the lifetime benefit of marriage into account when deciding to marry today, a small increase in costs via, say, proof of age requirement should not substantially alter this decision. Moreover, such costs should have a smaller effect if people expect to pay this cost later.

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